

Symmetry-Informed MARL: A Decentralized and Cooperative UAV Swarm Control Approach for Communication Coverage

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Abstract—Uncrewed aerial vehicle-mounted base stations (UAV-MBSs) provide flexible wireless connectivity, extending communication coverage in underserved areas. Recently, multi-agent reinforcement learning (MARL) has shown great potential for cooperative UAV swarm control to support efficient communication coverage in dynamic and complex environments. However, existing MARL-based methods often suffer from low sample efficiency due to its trial-and-error training characteristics, limiting its ability to control large UAV swarms with continuous state-action space and partial observation. We notice that UAV swarm systems in communication coverage tasks exhibit a spatial symmetry property, e.g., a rotation in the spatial observation of a UAV results in a same rotation in its optimal action. Exploiting this property, we formulate the task as a symmetric decentralized partially observable Markov decision process and introduce symmetry-informed MARL, featuring a novel network called the symmetry-informed graph neural network (SiGNN) to serve as the policy/value networks. SiGNN leverages the inherent symmetry in multi-UAV systems by embedding the symmetry into the network structure, thereby enhancing the training efficiency to handle large swarms with continuous control. Theoretical analysis shows that the SiGNN strictly preserves symmetry properties, which guarantees the effectiveness of the approach. Experiments in simulation were conducted to handle communication coverage using up to 20 UAVs with continuous control. Experimental results demonstrate that SiGNN-based MARL outperforms advanced baselines, verifying its superior sample efficiency, scalability and robustness.

Index Terms—Reinforcement learning, graph neural network, symmetry, UAV swarm, communication coverage.

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I. INTRODUCTION

UNCREWED aerial vehicle (UAV) swarms, enabled by advanced technologies such as artificial intelligence and mobile computing, have exhibited great potential in a variety of applications, ranging from civilian to military domains [1]. In particular, UAV-mounted base stations (UAV-MBSs) employ UAV to provide flexible and rapid-deployment wireless communication coverage for ground users and points of interest (PoIs) [2], [3]. Thanks to fast development of UAV technologies [4], [5], swarms of UAV-MBSs are able to operate cooperatively for communication coverage to provide high quality of service (QoS) in underserved areas such as disaster region, battlefields and congested places [6], [7], [8], [9]. As the demand for UAV swarms grows, it becomes crucial to develop robust control strategies that can enhance their operational capabilities and reliability in terms of coverage rate, energy and scalability.

Some existing studies for UAV swarm-based communication coverage focus on non-learning-based methods such as heuristic methods, approximation algorithms, and rule-based optimizations [10], [11], [12], [13]. These approaches rely extensively on human expertise and iterative solving process, limiting their practicability to real-time deployment and control in complex and dynamic environments. A promising approach to overcoming these limitations involves utilizing learning-based methods, such as multi-agent reinforcement learning (MARL). In recent years, MARL has shown a considerable promise for UAV swarm-based communication coverage, training deep policy networks to control UAVs in a decentralized and cooperative manner [14], [15], [16], [17], [18], [19]. Despite their demonstrated success, MARL-based methods often suffer from low sample efficiency due to its trial-and-error training characteristics, limiting its ability to control large UAV swarms in continuous state-action space and under partial observation, instead of operating in small number of UAVs [14], [15], [16], [18], simplified discrete space [17] or under global observation [14], [19].

Inspired by physics-informed machine learning, an emerging paradigm that seamlessly integrates data-driven learning models with physics [20], this paper explores the possibility of integrating prior knowledge (e.g., the physical properties of the environment and multi-UAV systems) into MARL. This integration aims to facilitate MARL-based UAV control to address the practical and challenging scenarios.

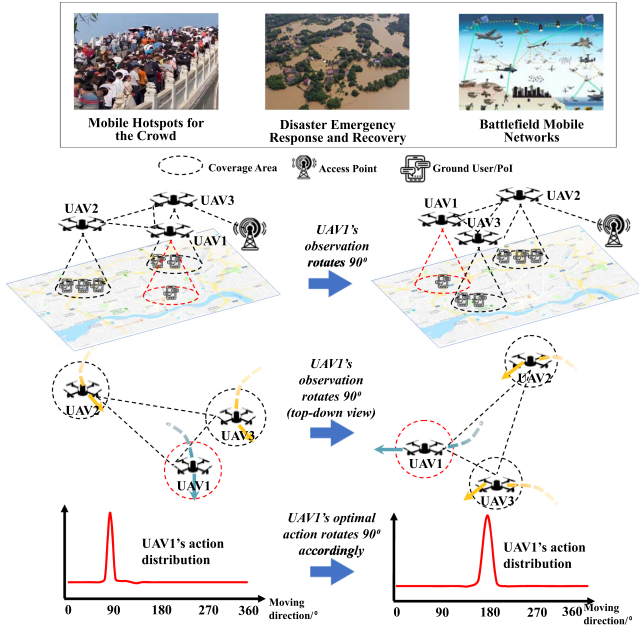


Fig. 1. Illustration of UAV swarm communication coverage tasks in underserved areas. The task generally exhibits spatial symmetry property, i.e., rotating a UAV's spatial observation by 90° leads to the same 90° rotation in the UAV's optimal action.

As shown in Fig. 1, we observe that UAV swarm systems engaged in communication coverage tasks often exhibit a property known as spatial symmetry. For example, a rotation in the spatial observation of a UAV results in a corresponding rotation in its optimal action. Symmetry in UAV swarm systems facilitates a more uniform and predictable behavior among individual UAVs, which is crucial for maintaining formation integrity and operational coherence in dynamic environments. This uniformity allows the swarm to respond collectively and efficiently to environmental variables, which enhances the system's overall stability and reliability.

By leveraging the symmetry property, we formulate the task as a symmetric decentralized partially observable Markov decision process (symmetric Dec-POMDP) and introduce a novel network structure called the symmetry-informed graph neural network (SiGNN), which serves as the backbone of policy and value network in the MARL framework. SiGNN leverages the inherent symmetry in multi-UAV systems by integrating this symmetry into its network structure, thus improving training and sample efficiency for coordinating large swarms with continuous control. Additionally, SiGNN includes an adjacency processing module that utilizes aggregation embedding to adapt to varying numbers of adjacent UAVs and PoIs under partial observation. Theoretical analysis shows that our network structure strictly preserves the symmetry, thereby providing a theoretical guarantee of the method's effectiveness.

The main contributions are as follows:

- Introduce the concept of spatial symmetry into the UAV swarm-based communication coverage task and newly formulate the task as a symmetric Dec-POMDP;
- Propose the SiGNN and the corresponding MARL, which integrates the symmetry inherently to the network structure for improving training efficiency and performance;

- Theoretically analyze that the proposed network structure strictly preserves the symmetry property, showing that our design is theoretically sound; and
- Design and conduct evaluation experiments using up to 20 UAVs with continuous control. Comprehensive comparison against advanced baselines are made to demonstrate our methods, in terms of coverage rate, energy consumption, coverage redundancy, and other performance on scalability and robustness.

II. RELATED WORKS

A. UAV Swarms for Coverage Problem

The area/communication coverage problem can be solved in part by UAV swarm techniques, which involves UAV scheduling [21], [22], path planning [23], [24], navigation control [25], [26], and other related tasks [27], [28]. Approaches in the literature can be broadly categorized into two main groups: non-learning-based and learning-based methods. Because there exist comprehensive surveys on this topic, we narrow our discussion to heuristic, rule-based and approximation techniques for the former, and multi-agent reinforcement learning for the latter. For more comprehensive surveys, one may refer to [1] and [29].

1) *Heuristic, Rule-Based and Approximation Techniques:* The coverage problem is usually formulated as an optimization problem. Related literature on solving the area and communication coverage with UAVs reveals a rich diversity of methods aimed at optimizing resource deployment. Heuristic methods such as the artificial bee colony (ABC) algorithm have been effectively used to minimize the number of UAVs required while ensuring optimal coverage and QoS [30]. As a combined method, the ordered ABC-based placement (OAP) algorithm clusters users and determines the optimal 3D positions of UAVs [12], [31]. Separate heuristics can be operated in a hierarchical manner to reduce computational time [32] while optimizing coverage efficiency for crowded areas, such as stadium environments [10].

Genetic algorithms (GA) and particle swarm optimization (PSO)-based methods are also widely used. For example, a multi-population GA method was introduced to enhance the optimization of a UAV's 2D position, aiming to maximize the number of users that the UAV can serve [33]. PSO has been leveraged to adjust UAV flight paths by simulating the social behavior of bird flocks to maximize area coverage and minimize resource usage for rescue operation [34]. An improved multi-objective particle swarm optimization (MOPSO) algorithm based on the adaptive angle area division was proposed to concern the threat constraint for improved multi-UAV task assignment [35].

Rule-based techniques are proposed to guide the scheduling of vehicle-mounted systems to optimize the coverage [36], [37]. For example, the edge-prior placement (EPP) for selecting users based on UAVs' proximity to the boundary of uncovered areas [38]. Rule-based methods are simple and fast to implement, albeit less adaptive to dynamic changes in the environments [13].

To handle more practical, constrained and scalable scenarios, approximation algorithms are introduced, especially for fair communications in UAV networks [6], [39], [40]. These approaches involve iterative solving processes and have to draw

a balance between processing time and optimality, limiting their practicability to large-scale and real-time deployment in complex and dynamic environments.

2) *MARL-Based Swarm Techniques*: UAV swarms function as multi-agent systems, and advanced multi-agent approaches, such as MARL,¹ are gaining increasing attention, particularly for large-scale control and highly dynamic coverage tasks. The most straightforward MARL-based manner to control UAV swarms at scale is the use of decentralized training with decentralized execution (DTDE) paradigm, in which agents are trained and executed independently without shared information. Independent proximal policy optimization (IPPO) has been applied to UAV-assisted fair communications [19], and techniques like reward shaping and parameter sharing were used to obtain multi-UAV coordination and high scalability (e.g., up to 20 UAVs). [17] refrained the use of global information in the training phase and applied recurrent graph network to aggregate multi-hop neighbors for inter-UAV collaboration. However, a complete independent training is impractical to achieve cooperative swarm control, and these works either more or less make use of global information in training, such as all UAVs' and ground users' positions [19], or simplify the task setting to operate in discrete state and action spaces [17].

The mainstream MARL framework is the centralized training with decentralized execution (CTDE), which trains agents using a shared, centralized information structure but executes them independently [41]. This is also the main MARL-based framework for multi-UAV coverage problems, offering improved training stability and coordination capabilities, which is of great importance when addressing practical multi-UAV control with continuous state and action spaces. Advanced CTDE MARL methods for UAV swarm coverage tasks involve the use of transformer MARL (T-MARL) for variable input dimensions [18], state-based game with actor-critic algorithm (SBG-AC) which features modeling the interactions among UAVs using game theory [42], hypergraph convolution mix DDPG for improved coordination in UAV deployment [43], graph convolution MARL [15] and state-action spaces [15], [44]. In addition, Xu et al. proposed a MARL algorithm that combines intrinsic reward based on feedback from large language models (LLM) and a grouped parameter sharing mechanism to effectively improve network coverage [45].

Unfortunately, these methods do not yet scale well for controlling even a relatively small number of UAVs (3 to 10 UAVs), indicating that controlling a large number of UAVs for communication coverage tasks in continuous spaces remains a challenging task.

B. Knowledge-Informed MARL

A major difficulty preventing MARL from scaling effectively for practical tasks lies in its low sample efficiency, due to the MARL's trial-and-error training characteristics. Extensive research has been conducted on improving the training

efficiency of MARL [46], [47], [48], [49]. With the advent of physics-informed machine learning [20], it is believed that incorporating external domain knowledge into learning models can enhance sample and training efficiency. Recent research has been conducted to improve machine learning models with domain knowledge, such as embedding the traffic models, seismic wave equations and fluid mechanics into neural networks for prediction and estimation purpose [50], [51], [52], [53].

C. Progress of Symmetry Property-Related Studies

Symmetry is a general property exhibited in the UAV swarm systems and can be leveraged to improve the MARL-based UAV control. Although still relatively scarce, symmetry-guided learning is emerging as a frontier of the deep learning field [54]. Related works include symmetry-guided data augmentation, which creates extra data by applying permutation transformations to homogeneous agents [55]. Symmetry consistency loss was proposed to leverage symmetry to regularize the MARL training [48], while this method has been extended to partial symmetric environments [49]. These methods learn the symmetry in a soft constrained manner, and heavy training overhead and hyperparameter tuning are involved though the requirement for the absolute trial-and-error sample data is relaxed. Hard symmetry constraint can encode symmetry into networks to decompose global symmetries into local symmetries that each agent can handle, while this method is reported suitable for discrete spaces [47].

Permutation symmetry is a type of non-geometric symmetry that describes a system's equivariance when its inputs or components are rearranged (permuted) in different orders. Recent studies have encoded permutation symmetry into policy networks to boost MARL [56]. Such permutation equivariant policy networks have been applied to the UAV communication scheduling of ground users, where the UAV trajectory design is boosted by data augmentation by exploiting rotational and reflection symmetries [57]. However, the exploration of spatial symmetry networks in UAV swarm tasks remains lacking which constitutes the primary contribution of this paper.

To our knowledge, this paper is the first to leverage the spatial symmetry-informed networks to the UAV swarm coverage tasks using MARL. The proposed SiGNN leads to an efficient symmetry-informed MARL framework, suitable for more practical and challenging settings. The paper will simultaneously concern continuous control space, partial observation (varying numbers of adjacent UAVs and PoIs) and a large number of UAVs (up to 20 UAVs), verifying our method's superior sample efficiency, scalability and robustness.

III. SYSTEM MODEL AND RESEARCH OBJECTIVE

A. System Model of the UAV Swarm

In this paper, we describe the UAV swarm control scenario for communication coverage on a 2D plane, as shown in Fig. 1. Let $\mathcal{N} = \{i|i = 1, 2, \dots, N\}$ be the set of UAVs equipped with base stations cooperating to support the communication needs of ground users and other PoIs. The plane

¹The term specifies deep learning-based MARL in this paper, which might be referred to as MADRL in some other papers.

is a continuous target area: $\{(x, y) | 0 \leq x \leq L_{\text{length}}, 0 \leq y \leq L_{\text{width}}; L_{\text{length}}, L_{\text{width}} \in \mathbb{R}^+\}$. The PoIs in the PoI set, $\mathcal{K} = \{k | k = 1, 2, \dots, K\}$, are randomly distributed across the target area. Each UAV launches from a random location and performs the communication coverage task over T time steps (each time step is indexed by t). Each UAV has a fixed coverage radius R_{cov} and observation radius R_{obs} on the 2D plane. We usually have $R_{\text{obs}} \geq R_{\text{cov}}$.

1) *UAV Model*: In this scenario, each UAV i is located at position (x_i, y_i) . The dynamics of the UAVs are driven by the velocities in the horizontal and vertical directions, denoted as $v_i = (v_{x_i}, v_{y_i})$ with $\|v_i\| \leq v_{\text{max}}$. Consequently, the UAV system adheres to the kinematic model as follows:

$$\begin{cases} \dot{x}_i = v_{x_i}, \\ \dot{y}_i = v_{y_i}. \end{cases} \quad (1)$$

This dynamics of UAVs is implemented as $x_i^{t+1} = x_i^t + v_{x_i}^t$ and $y_i^{t+1} = y_i^t + v_{y_i}^t$ in the simulation engine, where the superscript indicates the specific time step t . The controller of each UAV determines the velocity to be executed, i.e., the action of UAV i at time t is given as

$$a_i^t = (v_{x_i}^t, v_{y_i}^t). \quad (2)$$

From the viewpoint of UAV i , global observation is not available. Instead, a local observation o_i^t is accessible, consisting of four elements: 1) The position element $\mathbf{o}_i^{\text{pos}} = (x_i^t, y_i^t)$ containing the location information of the UAV itself, 2) the velocity element $\mathbf{o}_i^{\text{vel}} = (v_{x_i}^{t-1}, v_{y_i}^{t-1}) = a_i^{t-1}$, 3) the element $\mathbf{o}_i^{\text{UAV}}$ regarding the adjacent UAVs within the observable region, and 4) the element $\mathbf{o}_i^{\text{PoI}}$ regarding the adjacent PoIs within the coverage region.

Specifically, a UAV j is included into the adjacency set of UAV i if the relative distance $d_{ij} \leq R_{\text{obs}}$. Thus, we can define the adjacent element as $\mathbf{o}_i^{\text{UAV}} = \{(d_{ij}, \sin \theta_{ij}, \cos \theta_{ij}) | j \in \mathcal{N}_{/i}, d_{ij} \leq R_{\text{obs}}\}$, where θ_{ij} is the angle between the line segment from UAV i to UAV j and the heading direction of UAV i . Note, $\mathcal{N}_{/i}$ is the set with element i excluded. Similarly, the PoI related element is defined on the adjacent PoIs, i.e., $\mathbf{o}_i^{\text{PoI}} = \{(d_{ik}, \sin \theta_{ik}, \cos \theta_{ik}) | k \in \mathcal{K}, d_{ik} \leq R_{\text{cov}}\}$. As a result, the local observation of UAV i is represented as:

$$o_i^t = \{\mathbf{o}_i^{\text{pos}}, \mathbf{o}_i^{\text{vel}}, \mathbf{o}_i^{\text{UAV}}, \mathbf{o}_i^{\text{PoI}}\}. \quad (3)$$

The PoIs within the coverage region of one of the UAVs will be automatically served. To encourage cooperation among UAV swarm, it is assumed that each UAV can share their position information with its adjacent UAVs. We use the adjacency matrix $E \in \{0, 1\}^{N \times N}$ to indicate the connection between UAVs and it is a symmetric matrix.

2) *Evaluation Metrics*: An important part of the UAV swarm model is the evaluation metric system. Similar to [15], we introduce three key metrics to comprehensively evaluate the coverage effectiveness, distribution rationality and energy efficiency of the UAV swarm control in the tasks. The first metric is *coverage index*, measuring the average number of time steps that each PoI is covered and served by any UAV in one episode with T time steps. To normalize this value into $[0, 1]$, we further divide

it by the number of time steps past, i.e., T . At any time step t , a PoI k is indicated as covered if its distance to the closest the UAV i is smaller than the coverage radius R_{cov} . Therefore, the normalized coverage index c_{coverage} of an episode with T time steps is defined as the follow:

$$c_{\text{coverage}} = \frac{\sum_{t=1}^T \sum_{k=1}^K \mathbb{1}_{\text{cov}}^t(k)}{KT}, \quad (4)$$

where $\mathbb{1}_{\text{cov}}^t(k)$ is the indicator function $\mathbb{1}(\exists i \in \mathcal{N}, s.t., d_{ik} \leq R_{\text{cov}})$, indicating whether the PoI k is covered and served at the time step t . An indicator function $\mathbb{1}(\cdot)$ returns one if a specified condition is met, and zero otherwise. This index guarantees $c_{\text{coverage}} \in [0, 1]$ and evaluates how often each PoI is connected to one of the UAVs, quantifying the coverage performance of the UAV swarm in one episode.

However, the coverage index may score equally high as long as a fixed majority of PoIs are covered, and several UAVs may overlap to serve the same PoI. This is discouraged because, in emergency response scenarios such as flooding, it is crucial that each PoI is evenly served by the UAV swarm to mitigate situations where a large number of victims gather in the same place and a single UAV struggles to support the overwhelming communication demands. Here, we introduce the following *anti-overlap index*:

$$c_{\text{overlap}} = 1 - \frac{\sum_{t=1}^T \sum_{i=1}^N \mathbb{1}_{\text{overlap}}^t(i)}{NT}, \quad (5)$$

where $\mathbb{1}_{\text{overlap}}^t(i)$ is the indicator function $\mathbb{1}(\text{count}\{j \in \mathcal{N}_{/i} | d_{ij} \leq R_{\text{cov}}\} \geq n_{\text{thrd}})$, regarding whether the UAV i overlaps with more than n_{thrd} other UAVs. Here, we set the overlapping threshold $n_{\text{thrd}} = 3$. Obviously, we have $c_{\text{overlap}} \in [0, 1]$ and a larger c_{overlap} implies a more fair spatial distribution of UAVs when serving the PoIs in the area.

Finally, we consider the energy consumption of the UAV swarm to construct the third index. Following the power consumption model in [58], we decompose energy consumption of the UAV i into three parts when executing an action a_i^t at time t : 1) the base energy used to maintain the UAV's basic functioning status, which is a constant $e_0 > 1$, 2) the energy consumed to overcome air resistance due to flight speed, which is proportional to $\|v_i^t\|$, and 3) the additional energy used due to acceleration (changes in speed), which is proportional to $\|v_i^t - v_i^{t-1}\|$. It is noted that the power to support auxiliary communication is another part of the UAV energy consumption. However, this consumed energy is often smaller compared with the movement energy [42], especially when limited UAV communication resources are distributed to each PoI for light weight data transmission in emergent scenarios, such as disaster relief. Therefore, we focus on energy consumption for UAV movements and ignore the consumption for auxiliary communication for simplification. Specifically, the energy consumption of UAV i at time t is:

$$e_i^t = e_0 + \beta_{\text{speed}} \times \|v_i^t\| + \beta_{\text{acc}} \times \|v_i^t - v_i^{t-1}\|, \quad (6)$$

where β_{speed} and $\beta_{\text{acc}} \in \mathbb{R}^+$ are the coefficients converting the speed and acceleration to energy consumption, respectively. Because a UAV has a maximal speed v_{max} , the energy consumption rate is bounded by $e_{\text{max}} = e_0 + \beta_{\text{speed}} \|v_{\text{max}}\| + \beta_{\text{acc}} \|v_{\text{max}} - 0\|$.

For an undertrained UAV, its behavior could be irrational, leading to aimless flying around, which results in very high energy consumption. However, such extreme cases are rare as the training continues, and we are more concerned with how to distinguish the energy consumption between optimal and suboptimal strategies. Therefore, we leverage the logarithmic scaling to make extreme energy consumption less significant. The *energy index* is then defined as the follow:

$$c_{\text{energy}} = \log_{e_{\max}} \frac{\sum_{t=1}^T \sum_{i=1}^N e_i^t}{NT}. \quad (7)$$

Because $e_i^t \in (1, e_{\max}]$, the index is normalized to $(0, 1]$. The closer the energy index c_{energy} is to one, the more severe the energy consumption becomes.

We combine all the three indexes to define the comprehensive evaluation index, i.e. the *coverage anti-overlap energy (CAE) score* of the UAV swarm control performance in an episode, defined as:

$$CAE = \frac{c_{\text{coverage}} \times c_{\text{overlap}}}{c_{\text{energy}}}. \quad (8)$$

3) *Research Objective*: The overall research objective the paper concerned is to find an optimal UAV swarm control policy π^* that can maximize the expected CAE score in an episode, i.e.,

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} [CAE], \quad (9)$$

where $\mathbb{E}_{\pi} [CAE]$ is the episodic expectation of the CAE score when the UAV swarm is controlled under the policy π . For a UAV swarm system, the policy π is composed of individual policies π_i , i.e., $\pi = (\pi_1, \dots, \pi_N)$. Based on the CAE score, a more reasonable policy should obtain a higher coverage index while mitigating overlapping situations and energy consumption.

B. Preliminaries of Symmetric Dec-POMDP

1) *Definition of Dec-POMDP*: We first provide the definition of the decentralized partially observable Markov decision process (Dec-POMDP) [48].

An N -agent cooperative task can be formulated as an Dec-POMDP, represented using a tuple $(\mathcal{N}, \mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{T}, r, \gamma, \Omega)$. Here, $\mathcal{N}$ denotes the set of agents, $\mathcal{S}$ is the state space, $\mathcal{O}$ is the joint observation space, and $\mathcal{A} = \mathcal{A}_1 \times \dots \times \mathcal{A}_N$ is the joint action space consisting of the individual action space $\mathcal{A}_i$ of each agent $i \in \mathcal{N}$. The transition function $\mathcal{T} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ describes the underlying dynamics of the environment. The (joint) reward function $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ can be predefined and shaped to adapt to the task characteristics. The joint return is defined by $\sum_{t=0}^{\infty} (\gamma)^t r(s^t, a^t)$, where $\gamma \in [0, 1]$ is the discount factor. Each agent i has an individual observation space $\mathcal{O}_i$ and a corresponding observation function $\Omega_i : \mathcal{S} \rightarrow \mathcal{O}_i$. At state s^t , each agent i receives a partial observation $o_i^t = \Omega_i(s^t) \in \mathcal{O}_i$. The joint observation is denoted as $o^t = (o_1^t, \dots, o_N^t) \in \mathcal{O}$, with the observation function set $\Omega = (\Omega_1, \dots, \Omega_N)$. Each agent i follows its individual policy $\pi_i : \mathcal{O}_i \times \mathcal{A}_i \rightarrow [0, 1]$. The policy takes the partial observation o_i^t as input and generates an action distribution, from which an executable action is sampled, i.e., $a_i^t \sim \pi_i(\cdot | o_i^t)$. For a deterministic policy, $a_i^t = \pi_i(o_i^t)$. A MARL algorithm aims to learn an optimal joint policy $\pi^* =$

$(\pi_1^*, \dots, \pi_N^*)$, enabling all agents to cooperate to maximize the expected joint return, defined as $\mathbb{E}_{\pi} [\sum_{t=0}^{\infty} (\gamma)^t r(s^t, a^t)]$.

2) *Definition of Mathematical Symmetry*: The mathematical symmetry is a property that an object, such as state and geometric shape, remain unchanged after a transformation. To fully define this property, it is necessary to provide an overview of the concepts of groups and transformations. A *group* G in math is a set with a *binary operator* that satisfies the four mathematical properties (also known as group axioms): identity, inverse, closure, and associativity. According to the characteristics of the UAV swarm-based coverage task, this paper primarily lies its interest in the group “E(2)”, which consists of group actions such as translations, rotations, reflections, and their finite combinations in 2D.

A group action is also called a *transformation*. For a function $f : \mathcal{X} \rightarrow \mathcal{Y}$, if there exist two transformation operators $L_g : \mathcal{X} \rightarrow \mathcal{X}$ and $K_g : \mathcal{Y} \rightarrow \mathcal{Y}$ associated with a group element $g \in G$, such that $K_g[f(x)] = f(L_g[x])$ for any $x \in \mathcal{X}$, then we say that the function f is *equivariantly symmetric* under the transformations L_g and K_g . When $K_g = I$, the identity function, then we say that the function f is *invariantly symmetric* under the transformation L_g . The equivariant symmetry and the invariant symmetry constitutes the concepts of symmetry in this paper.

3) *Definition of Symmetric Dec-POMDP*: The symmetric Dec-POMDP is a subclass of Dec-POMDP, of which the optimal policy possesses the symmetry property. Specifically, let $\pi^* = (\pi_1^*, \dots, \pi_N^*)$ be the optimal joint policy following which the agents can operate cooperatively to maximize the expected joint return. Then, we call the Dec-POMDP a *symmetric Dec-POMDP* if π^* is equivariantly-symmetric, i.e.,

$$\pi_i^*(L_g[o_i]) = K_g[\pi_i^*(o_i)] = K_g[a_i^*], \forall i \in \mathcal{N}, \quad (10)$$

under some transformations $L_g : \mathcal{O} \rightarrow \mathcal{O}$ and $K_g : \mathcal{A} \rightarrow \mathcal{A}$ on the observation space and action space, respectively.

C. Modeling UAV Coverage Task as Symmetric Dec-POMDP

The physical meaning of “symmetry” in a UAV swarm system defined in this paper is that when the UAV’s physical observation features are altered by a specific transformation L_g , the UAV’s control action should change through a corresponding transformation K_g to maintain consistent performance. Fig. 2 provides a simple yet highly practical transformation operation—rotation.

Specifically, the rotation transformation through an angle θ in a 2D system around a specific point is written as a matrix $R_{\theta} = [\cos\theta, -\sin\theta; \sin\theta, \cos\theta]$, and the set of R_{θ} is called a rotation group g . By operating the rotation on the observation vector $\mathbf{x}$ of a UAV in the swarm, we are conducting the calculation: $L_g[\mathbf{x}] = R_{\theta}\mathbf{x} = \mathbf{x}'$. In the original state depicted on the left, the optimal policy π directs UAV1 to move downward to avoid overlapping coverage with UAV3, i.e., $\mathbf{a} = \pi(\mathbf{x})$. When the whole state, encompassing the features of UAV swarm systems and PoIs, undergoes a rotation L_g with $\theta = 90^\circ$, the UAV1’s observation rotates 90° , accordingly, and the optimal action for UAV1 becomes moving leftward, resulting in the same 90° rotation in UAV1’s action. Because we have $\mathbf{a}' = \pi(R_{\theta}\mathbf{x}) =$

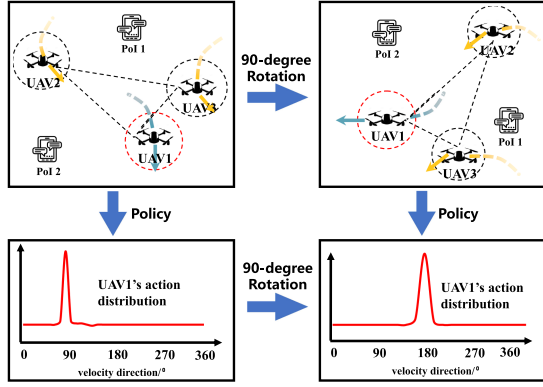


Fig. 2. Illustration of rotation symmetry in UAV swarm communication coverage tasks.

$\pi(L_g[\mathbf{x}]) = L_g[\pi(\mathbf{x})] = L_g[\mathbf{a}]$, by setting $K_g = L_g$, the policy π is equivariantly symmetric to the rotation transformation R_θ .

In practical applications such as communication coverage and drone arts, the symmetry property typically holds. Particularly when UAV swarms perform missions based on symmetrical structures, e.g., formation patterns (circles, hexagons, or grids), the symmetry property is inherent to the tasks. In such cases, leveraging the symmetry as prior knowledge to train the policy can reduce the need for additional learning efforts, improving the training and mission execution efficiency.

Without special notice, we focus on rotational transformations in this paper, although several other transformation operations such as reflections and translations can also apply. Please note that, in this paper, the rotation operation transforms only the states of UAVs and the locations of PoIs. The rotation of other environmental positions is irrelevant as they are not part of the observation.

Based on the symmetry property, we can model the UAV swarm control task for communication coverage as a symmetric Dec-POMDP. We set the discounted factor γ to 0.99 and use the kinematic model in (1) as the core of the transition function $\mathcal{T}$ to generate the state of the next time step. To fully formulate the symmetric Dec-POMDP, we define the state/observation, action and reward, accordingly.

1) *State Space $\mathcal{S}$ and Observation Space $\mathcal{O}$* : The state space $\mathcal{S}$ includes the positions of all the UAVs and PoIs on the 2D area as well as all the velocity information of UAVs. The observation function Ω_i maps the state s^t into the individual observation o_i^t of UAV i as defined in (3). The set of all feasible joint observation defines the observation space $\mathcal{O}$.

2) *Action Space $\mathcal{A}$* : The action space $\mathcal{A}$ jointly includes all the individual velocity spaces $\mathcal{A}_i$. A feasible action a_i^t is defined by (2).

3) *Reward Function r* : The remaining element is the reward function r . Although we can directly use the CAE score in (8) as the reward, it is impractical to the partially observable settings. Also, it can be difficult to determine the contribution each agent made when directly using the CAE score. Designing individual rewards can clarify the impact of each agent's actions, making it easier to train and optimize their behaviors. Furthermore, the

CAE score requires post-processing, including averaging and normalization, which can obfuscate the distinction between sub-optimal actions, making it challenging to train a satisfactory policy. Here, we design the individual reward function r_i^t , which consists of three terms: *coverage term*, *energy term* and *overlap term*.

The coverage term calculates the independent coverage reward for each agent:

$$r_i^{\text{cov}} = \lambda^{\text{cov}} \times \frac{|\mathbf{o}_i^{\text{Pol}}|}{|\mathcal{N}| \times |\mathcal{K}|}, \quad (11)$$

where $|\mathbf{o}_i^{\text{Pol}}|$ counts the number of PoIs covered and served by UAV i as defined in (3). This value encourages the UAV to serve the area with dense PoIs. The more PoIs each UAV covers, the more likely that each PoI k can be served, i.e., the probability of $\mathbb{1}_{\text{cov}}^t(k) = 1$ at each time slot is large. And this contributes to a larger coverage index c_{coverage} in (4). The energy term penalizes the energy consumption:

$$r_i^{\text{energy}} = -\lambda_1^{\text{eg}} \times \|a_i^t\| - \lambda_2^{\text{eg}} \times \|v_i^t\|. \quad (12)$$

Obviously, it is a direct modification with a negative sign of the individual energy consumption given in (6), and is negatively contributing to the energy index c_{energy} in (7). The overlap term is to prevent UAVs from overly concentrating in the same area, ensure reasonable distribution, and avoid redundant coverage:

$$r_i^{\text{overlap}} = -\lambda^{\text{ol}} \times \mathbb{1}_{\text{overlap}}^t(i), \quad (13)$$

where each UAV can observe its local coverage redundancy with other UAVs, indicated using $\mathbb{1}_{\text{overlap}}^t(i)$, which is an individual version of the anti-overlap index c_{overlap} in (5). λ^{cov} , λ_1^{eg} , λ_2^{eg} , λ^{ol} are coefficients in $\mathbb{R}^+$. Taking all the three terms into account, the individual reward function is defined as

$$r_i^t = r_i^{\text{cov}} + r_i^{\text{energy}} + r_i^{\text{overlap}}. \quad (14)$$

Thus far, the UAV swarm control task for communication coverage has been formulated as a symmetric Dec-POMDP model. Here, the joint return becomes $\sum_{i=1}^N \sum_{t=1}^T (\gamma)^t r_i^t$, which is approximately proportional to the CAE score of an episode. The goal of a MARL method is to find a UAV swarm control policy π to maximize the expected joint return.

IV. PROPOSED SYMMETRY-INFORMED GRAPH NEURAL NETWORK BASED APPROACH

This section introduces the proposed SiGNN framework in Fig. 3 and provides its design details. We also presents the MARL training framework based on SiGNN. As a main contribution of the paper, the proposed SiGNN-based networks automatically preserve the rotation symmetry property, which is an intrinsic property in symmetric Dec-POMDP. These symmetry-preserving networks positively impact the RL process by eliminating the need for additional trial-and-error to learn a symmetry-preserving policy. This allows the policy to better align with the symmetric Dec-POMDP during training.

For ease of describing the SiGNN, we further divide the features of UAV i 's individual observation $o_i^t =$

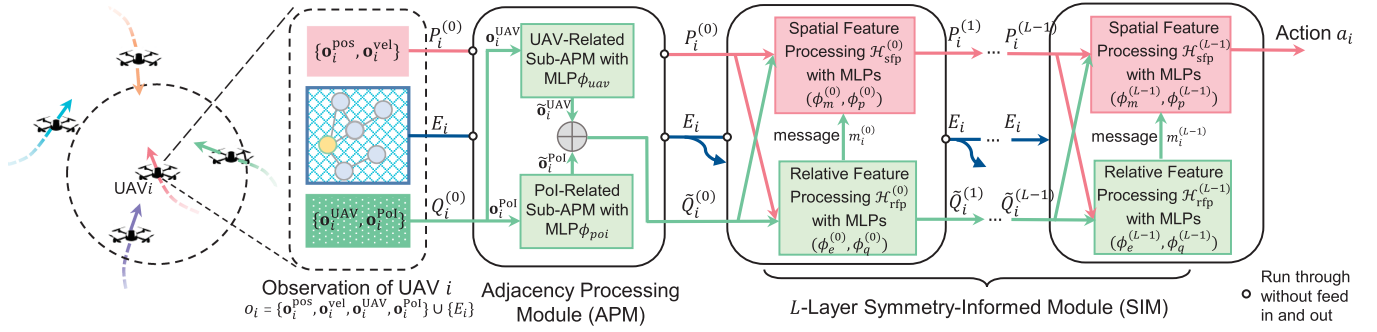


Fig. 3. Overview of the SIGNN framework, which constitutes the core network to train and control the behaviors of UAV i .

$\{\mathbf{o}_i^{\text{pos}}, \mathbf{o}_i^{\text{vel}}, \mathbf{o}_i^{\text{UAV}}, \mathbf{o}_i^{\text{PoI}}\}$ into two parts: the *absolute spatial features* P_i and the *ego-centered relative features* Q_i . The absolute spatial features $P_i = \{\mathbf{o}_i^{\text{pos}}, \mathbf{o}_i^{\text{vel}}\}$ includes features defined in the absolute coordinate system, i.e., the absolute position and velocity, which will change after rotation L_g , i.e., $P_i \neq L_g[P_i]$. The ego-centered relative features $Q_i = \{\mathbf{o}_i^{\text{UAV}}, \mathbf{o}_i^{\text{PoI}}\}$ includes features defined in the ego-centered coordinate system, i.e., the relative distances and direction angles of the adjacent UAVs and PoIs within the observable region. Obviously, Q_i is invariant to the rotation L_g , i.e., $Q_i = L_g[Q_i]$, because the relative distances and direction angles remain the same after rotation from the viewpoint of UAV i . Here, the time step symbol t is omitted for simplifying the representation.

Please note that the adjacency matrix E includes the information about which UAVs are observable and connectable by UAV i , such that their position information is shareable with each other. E_i denotes the i -th column of E .

A. SiGNN Framework Overview

As illustrated in Fig. 3, the overall architecture of SiGNN consists of two primary modules: the *Adjacency Processing Module (APM)* and the *Symmetry-Informed Module (SIM)*. The APM is designed to pre-process the ego-centered relative features Q_i . These features are unfriendly to policy networks because their dimensions keep changing as the adjacent UAVs and PoIs move in and out of UAV i 's observation. To address this issue, APM employs two sub-modules to pre-process the features $\mathbf{o}_i^{\text{UAV}}$ and $\mathbf{o}_i^{\text{PoI}}$ respectively, converting them to the *transformed* ego-centered relative features $\tilde{Q}_i$, an embedding of fixed dimension.

The SIM takes $\{P_i, \tilde{Q}_i, E_i\}$ as its input for further processing. It also contains two sub-modules to process the P_i and $\tilde{Q}_i$ jointly. An important note is that the outputs $\{\tilde{P}_i^{(1)}, \tilde{Q}_i^{(1)}\}$ are the transformed counterparts of the original $P_i, \tilde{Q}_i$, maintaining the same dimension and form. Then, the $\{\tilde{P}_i^{(1)}, \tilde{Q}_i^{(1)}, E_i\}$ are fed forward to the next SIM as inputs. There are L SIMs to construct a L -layer feed-forward network structure and we call it as *L-layer SIM*. Note that E_i consistently serves as part of the inputs of the SIM in each layer.

The SIM of layer $(L-1)$ releases the final output $\{\tilde{P}_i^{(L)}, \tilde{Q}_i^{(L)}\}$, and we formulate the whole process of SiGNN

parameterized by θ_i as:

$$\{P_i^{(L)}, \tilde{Q}_i^{(L)}\} = \text{SiGNN}(P_i, Q_i, E_i; \theta_i). \quad (15)$$

Here, we can extract $\mathbf{o}_i^{\text{vel}(L)}$ from $P_i^{(L)}$, the counterparts of the original velocity features $\mathbf{o}_i^{\text{vel}}$, as the action a_i . In this way, the SiGNN serves as a policy network of UAV i , i.e., $a_i = \pi_{\text{SiGNN}, i}(o_i, E_i; \theta_i)$, mapping the individual observation o_i to action a_i for operation in the current time step.

B. Adjacency Processing Module (APM)

The APM pre-processes the ego-centered relative features $Q_i = \{\mathbf{o}_i^{\text{UAV}}, \mathbf{o}_i^{\text{PoI}}\}$, which are rotation-invariant. Features Q_i include partially observational information from the viewpoint of UAV i , guiding its intention to cooperate with surrounding UAVs and serve the nearby PoIs for the best coverage. In contrast to global observation with fixed numbers of UAVs and PoIs, the dimension of Q_i keeps varying. The APM is able to deal with the varying inputs and aggregate the adjacency information by employing two adjacency processing sub-modules (sub-APMs): a *UAV-related sub-APM* and a *PoI-related sub-APM*.

1) *UAV-Related Sub-APM*: The UAV-related sub-APM processes $\mathbf{o}_i^{\text{UAV}}$, the features of adjacent UAVs. Let $\mathcal{N}(i)$ be the set of UAVs in the observable region, i.e., $j \in \mathcal{N}(i)$ if $d_{ij} \leq R_{\text{obs}}$. This sub-module leverages a multi-layer perceptron (MLP) ϕ_{uav} to calculate a weight for each adjacent UAV j and then makes weighted sum as its output $\tilde{\mathbf{o}}_i^{\text{UAV}}$. The MLP ϕ_{uav} has multiple hidden layers using rectified linear unit (ReLU) as the activation function. Without special notice, all the MLPs use this structural design. Specifically, this sub-module is formulated as

$$\tilde{\mathbf{o}}_i^{\text{UAV}} = \frac{1}{N} \sum_{j \in \mathcal{N}(i)} [\phi_{uav}(\mathbf{o}_{ij}^{\text{UAV}})] \cdot \mathbf{o}_{ij}^{\text{UAV}}, \quad (16)$$

where $\mathbf{o}_{ij}^{\text{UAV}}$ denotes the features of adjacent UAV j and N is the number of all UAVs for scaling the output. Because this aggregation is a linear combination of adjacent UAVs, the outputted $\tilde{\mathbf{o}}_i^{\text{UAV}}$ enjoys a fixed dimension and rotation invariance is ensured in this sub-module. The weights are sophisticatedly assigned by ϕ_{uav} , which is parameterized, enabling this sub-module to process the information of adjacent UAVs in a learnable manner.

2) *PoI-Related Sub-APM*: The PoI-related sub-APM processes $\mathbf{o}_i^{\text{PoI}}$, the features of adjacent PoIs. Let $\mathcal{K}(i)$ be the set

of PoIs in the coverage region, i.e., $k \in \mathcal{K}(i)$ if $d_{ik} \leq R_{\text{cov}}$. We equip this sub-module with MLP ϕ_{poi} for aggregation:

$$\tilde{\mathbf{o}}_i^{\text{Pol}} = \frac{1}{K} \sum_{k \in \mathcal{K}(i)} [\phi_{poi}(\mathbf{o}_{ik}^{\text{Pol}})] \cdot \mathbf{o}_{ik}^{\text{Pol}}, \quad (17)$$

where $\mathbf{o}_{ik}^{\text{Pol}}$ denotes the features of adjacent PoI k and K is the number of all PoIs for scaling the output. Obviously, this sub-module is also rotation-invariant.

3) *The Outputs of APM*: The outputs of the two sub-modules are then concatenated to form the aggregated embedding $\tilde{Q}_i = \{\tilde{\mathbf{o}}_i^{\text{UAV}}, \tilde{\mathbf{o}}_i^{\text{Pol}}\}$. The $\tilde{Q}_i$ released by the APM along with the absolute spatial features P_i and adjacency matrix E_i forms the input of the SIM.

There are L layers of SIMs in the SiGNN architecture, and we denote the input of the SIM in layer l as $\{P_i^{(l)}, \tilde{Q}_i^{(l)}, E_i\}$ to emphasize the layer-wise information using superscripts, which is initialized by setting $P_i^{(0)} = P_i$ and $\tilde{Q}_i^{(0)} = \tilde{Q}_i$. E_i is a constant matrix and superscript is unnecessary.

C. Symmetry-Informed Module (SIM)

The SIM in layer l jointly processes the *hidden* absolute spatial features $P_i^{(l)}$ and transformed ego-centered relative features $\tilde{Q}_i^{(l)}$ to compute $\{P_i^{(l+1)}, \tilde{Q}_i^{(l+1)}\}$ with same dimension, serving as part of the input to the next SIM layer.

A SIM also contains two sub-modules: a *spatial feature processing (SFP) sub-module* and a *relative feature processing (RFP) sub-module*, which are denoted as $\mathcal{H}_{\text{sfp}}^{(l)}$ and $\mathcal{H}_{\text{rfp}}^{(l)}$, respectively. The feed forward process is described as :

$$(\tilde{Q}_i^{(l+1)}, m_i^{(l)}) = \mathcal{H}_{\text{rfp}}^{(l)}(P_i^{(l)}, \tilde{Q}_i^{(l)}, E_i), \quad (18)$$

$$P_i^{(l+1)} = \mathcal{H}_{\text{sfp}}^{(l)}(P_i^{(l)}, \tilde{Q}_i^{(l)}, E_i, m_i^{(l)}). \quad (19)$$

Because $\mathcal{H}_{\text{sfp}}^{(l)}$ requires some information $m_i^{(l)}$ generated from $\mathcal{H}_{\text{rfp}}^{(l)}$ to continue its processing, we describe the $\mathcal{H}_{\text{rfp}}^{(l)}$ first.

1) *Relative Feature Processing $\mathcal{H}_{\text{rfp}}^{(l)}$* : The RFP sub-module $\mathcal{H}_{\text{rfp}}^{(l)}$ takes $\{P_i^{(l)}, \tilde{Q}_i^{(l)}, E_i\}$ as inputs and is equipped with two MLPs $\phi_e^{(l)}$ and $\phi_q^{(l)}$. We leverage $\phi_e^{(l)}$ to calculate the message $m_{ij}^{(l)}$ from each adjacent UAV j and merge all the adjacent messages to calculate the next $\tilde{Q}_i^{(l+1)}$ based on another MLP $\phi_q^{(l)}$. This technique is called *message-passing*, which is commonly used in graph network-based methods [54]. Specifically, the $\mathcal{H}_{\text{rfp}}^{(l)}$ is formulated as:

$$m_{ij}^{(l)} = \phi_e^{(l)}\left(\tilde{Q}_i^{(l)}, \tilde{Q}_j^{(l)}, \|P_i^{(l)} - P_j^{(l)}\|^2\right), \quad (20)$$

$$\tilde{Q}_i^{(l+1)} = \phi_q^{(l)}\left(\tilde{Q}_i^{(l)}, \sum_{j \in \mathcal{N}(i)} m_{ij}^{(l)}\right), \quad (21)$$

where the term $\|P_i^{(l)} - P_j^{(l)}\|^2$ gives the relative squared distance between two vectors. The set of messages received by UAV i from adjacent UAVs is denoted as $m_i^{(l)} = \{m_{ij}^{(l)} | j \in \mathcal{N}(i)\}$. $\mathcal{N}(i)$ is extracted from E_i , and this is why we make use of E_i

TABLE I
COMPUTATIONAL COMPLEXITY ANALYSIS OF SiGNN ON EACH UAV

Module/Sub-Module	Approximate MACCs
APM $\tilde{Q}_i^{(0)}$ in Eq.(16),Eq.(17)	$\xi_1 = d_{o_{ij}} \times d_{hid} \times (N_i + K_i)$
Message-passing m_{ij} in Eq.(20)	$\xi_2 = d_P + (2d_{\tilde{Q}} + 1) \times d_{hid} \times d_m$
Relative feature $\tilde{Q}_i^{(l+1)}$ in Eq.(21)	$\xi_3 = (d_{\tilde{Q}} + 1) \times d_{hid} \times d_{\tilde{Q}}$
Spatial feature $P_i^{(l+1)}$ in Eq.(22)	$\xi_4 = d_{\tilde{Q}} \times d_{hid} \times 1 \times d_P$ + $N_i \times [d_m \times d_{hid} \times 1 \times d_P]$

as part of input. Both (20) and (21) define the core operation of the RFP sub-module $\mathcal{H}_{\text{rfp}}^{(l)}$ in (18)

2) *Spatial Feature Processing $\mathcal{H}_{\text{sfp}}^{(l)}$* : With the messages $m_i^{(l)}$ of adjacent UAVs, the SFP sub-module $\mathcal{H}_{\text{sfp}}^{(l)}$ employs two MLPs $\phi_m^{(l)}$ and $\phi_p^{(l)}$ to update the spatial features, which is formulated as

$$P_i^{(l+1)} = \phi_p^{(l)}(\tilde{Q}_i^{(l)}) \cdot P_i^{(l)} + C \sum_{j \in \mathcal{N}(i)} \phi_m^{(l)}(m_{ij}^{(l)}) \cdot (P_i^{(l)} - P_j^{(l)}), \quad (22)$$

where C is a constant to control the scale of the second term, involving the influence from adjacent UAVs. (22) indicates that the spatial features $P_i^{(l)}$ is iteratively updated by a linear combination of itself and its differences from all the adjacent UAVs, with the weights calculated by $\phi_p^{(l)}$ and $\phi_m^{(l)}$, respectively. The weight from $\phi_p^{(l)}$ is determined by the transformed ego-centered relative features $\tilde{Q}_i^{(l)}$ to modulate the current $P_i^{(l)}$ and the weights from $\phi_m^{(l)}$ are mapped from the corresponding messages to regulate the relative features $(P_i^{(l)} - P_j^{(l)})$ from the nearby UAV j .

After L -layer feed forward SIM processing, we obtain the final output $\{P_i^{(L)}, \tilde{Q}_i^{(L)}\}$, in which $P_i^{(L)} = \{\mathbf{o}_i^{\text{pos}(L)}, \mathbf{o}_i^{\text{vel}(L)}\}$. The UAV i takes $\mathbf{o}_i^{\text{vel}(L)}$ as its action a_i at the current time step and the SiGNN has now been fully defined.

The multiply-accumulate operations (MACCs) of SiGNN on each UAV is presented in Table I. It can be seen that the MLP components, such as ϕ_q , contribute to the most computational complexity due to the involvement of multiply layers. The overall MACC of a SiGNN with L layers is: $\xi_1 + L \times [\xi_2 \times N_i + \xi_3 + \xi_4]$, which has the complexity of $O(LN_i d_m d_{hid} + K_i d_{hid})$.

D. Symmetry-Informed MARL Based on SiGNN

The SiGNN framework can serve not only as the policy network (actor) but also as the core component of the value network (critic) to support the centralized training with decentralized execution (CTDE) paradigm in MARL. In CTDE, UAVs are trained on a global and centralized critic, where information from all UAVs can be utilized to evaluate the behaviors. However, during execution, each UAV operates independently, using local observation and individual actor. It is straightforward to employ SiGNN as the actor. Parameter sharing technique can apply because UAVs are homogeneous to share the same

Algorithm 1: Procedure of Symmetry-Informed MARL.

Input: Number of UAVs N , episode length T , episode batch size M .

Output: Optimized SiGNN-based policy networks π_{SiGNN} .

- 1: Initialize $\theta = \{\theta_i\}_{i=1}^N$, the parameters of SiGNN-based actors π_{SiGNN} , and ψ , the parameters for SiGNN-based critic Ψ_{SiGNN} and replay buffer $\mathcal{D}$.
- 2: **for** each episode **do**
- 3: **for** each timestep t from 1 to T **do**
- 4: Get global observation o^t and adjacency matrix E^t .
- 5: **for** each agent i from 1 to N **do**
- 6: Get individual observation o_i^t and E_i^t .
- 7: Get action $a_i^t = \pi_{\text{SiGNN},i}(o_i^t, E_i^t; \theta_i)$ on (15).
- 8: Execute action a_i^t , get o_i^{t+1} from simulation (1) and receive r_i^t on (14).
- 9: Obtain an individual transition $(o_i^t, a_i^t, r_i^t, o_i^{t+1})$.
- 10: **end for**
- 11: Store global transition $\{(o_i^t, a_i^t, r_i^t, o_i^{t+1})\}_{i=1}^N$ into replay buffer $\mathcal{D}$.
- 12: **end for**
- 13: **if** episode $\bmod M = 0$ **then**
- 14: **for** each mini-batch **do**
- 15: Sample mini-batch $\mathcal{B} \sim \mathcal{D}$.
- 16: Update parameters θ of the actors π_{SiGNN} on $\mathcal{B}$.
- 17: Update parameters ψ of the critic Ψ_{SiGNN} on $\mathcal{B}$.
- 18: **end for**
- 19: **end if**
- 20: **end for**

individual actor. The remaining component is the centralized critic.

Let $P = (P_1, \dots, P_N)$, $Q = (Q_1, \dots, Q_N)$, and $o = \{P, Q\}$ be the global observation, and $J = [1]^{N \times N}$ be the all-one matrix denoting a fully observable condition. We introduce the following SiGNN-based centralized value network (critic) Ψ_{SiGNN} , parameterized by $\psi = \{\psi_1, \psi_2\}$:

$$V = \Psi_{\text{SiGNN}}(o, J; \psi) = \Psi_{\text{SiGNN}}(P, Q, J; \psi), \quad (23)$$

Specifically, the critic's calculation is broken down as follows:

$$\{P_i^{(L)}, \tilde{Q}_i^{(L)}\} = \text{SiGNN}(P_i, Q_i, J_i; \psi_1), \quad (24)$$

$$U = (U_1, \dots, U_N), U_i = \|\mathbf{o}_i^{\text{vel}(L)}\|^2, \quad (25)$$

$$V = \text{FC}(\tanh(U, Q); \psi_2), \quad (26)$$

where $\tanh(\cdot)$ is the element-wise hyperbolic function and $\text{FC}(\cdot)$ is the one-layer fully connected network. The designed critic makes an ample use of the velocity and the transformed ego-centred relative features to calculate the centralized value. The squared velocity counterparts in (25) are used to reflect the energy situation. The position counterparts are not used in (26) because it was found to have a trivial impact in practice.

The MARL that leverages the SiGNN-based actor and critic is called a *symmetry-informed MARL*. By default, we use the popular multi-agent proximal policy optimization (MAPPO) [59] as the backbone of the MARL,

though other CTDE MARL frameworks can also apply. The MAPPO learns policy by maximizing the objective: $\mathcal{J}_\pi(\theta) = \sum_{i=1}^N \mathbb{E}_{\pi_{\theta_{\text{old},i}}} \left[\frac{\pi_{\theta_i}(a_i^t|o_i^t)}{\pi_{\theta_{\text{old},i}}(a_i^t|o_i^t)} A_{\theta_{\text{old},i}}(o_i^t, a_i^t) \right]$, where π_{θ_i} denotes the $\pi_{\text{SiGNN},i}(\cdot; \theta_i)$ of agent i , which is the current policy to optimize. $\pi_{\theta_{\text{old},i}}$ is the behavior policy of trajectory experiences. $A_{\theta_{\text{old},i}}(o_i^t, a_i^t)$ denotes the advantage function, computed by Generalized Advantage Estimation (GAE) on the trajectory. The critic network is trained to maximize the objective: $\mathcal{J}_V(\psi) = -\mathbb{E}_\pi [V_{\text{target}}^\pi(o^t) - V_\psi^\pi(o^t)]^2$, where $V_{\text{target}}^\pi(o^t)$ denotes the target value (temporal difference target or reward-to-go target) for the global observation o^t and $V_\psi^\pi(o^t)$ denotes $\Psi_{\text{SiGNN}}(\cdot; \psi)$ to evaluate the current policy π . The actor and critic networks are updated using $\nabla_\theta \mathcal{J}_\pi$ and $\nabla_\psi \mathcal{J}_V$, respectively, with gradient clipping. Pseudocode for the symmetry-informed MARL is presented in Algorithm 1.

V. THEORETICAL ANALYSIS OF SYMMETRY IN SiGNN-BASED POLICY AND VALUE NETWORKS

This section theoretically shows that the proposed SiGNN-based policy and value networks are able to strictly preserve the symmetry property under the rotation transformation L_g . Specifically, we show that each SiGNN-based policy network $\pi_{\text{SiGNN},i}$ is equivariantly symmetric under the rotation transformation L_g , and the SiGNN-based value network Ψ_{SiGNN} is invariantly symmetric under L_g . The first claim regarding the SiGNN-based policy network states as:

Proposition 1: The SiGNN-based policy network $a_i = \pi_{\text{SiGNN},i}(o_i, E_i; \theta_i)$ is equivariantly symmetric under the rotation transformation L_g , which satisfies:

$$L_g[a_i] = \pi_{\text{SiGNN},i}(L_g[o_i], L_g[E_i]; \theta_i), \forall i \in \mathcal{N}. \quad (27)$$

Proof: Because the action $a_i = \mathbf{o}_i^{\text{vel}(L)}$ is extracted from $P_i^{(L)}$ in (15), which is part of the output of the SiGNN network, we can focus on analyzing the symmetry property of SiGNN. By imposing the rotation transformation L_g on SiGNN, we can make a stronger claim that

$$\{L_g[P_i^{(L)}], \tilde{Q}_i^{(L)}\} = \text{SiGNN}(L_g[\{P_i, Q_i, E_i\}]; \theta_i), \quad (28)$$

which is a sufficient condition to (27). Because the rotation does not change the relative features, the ego-centered relative features Q_i and the adjacency matrix E are rotation-invariant, i.e., $Q_i = L_g[Q_i]$ and $E_i = L_g[E_i]$.

The APM is the first module in SiGNN to convert the Q_i to the transformed embedding $\tilde{Q}_i$. $\tilde{Q}_i$ inherits the rotation invariance because it is a linear combination of features in Q_i . Therefore, we have $\tilde{Q}_i = L_g[\tilde{Q}_i]$. Then, the SIM in layer 0 takes $\{L_g[P_i^{(0)}], \tilde{Q}_i^{(0)}, E_i\}$ as inputs and feed the results to the SIM in the next layer.

In each layer l , the RFP sub-module $\mathcal{H}_{\text{rfp}}^{(l)}$ is invariantly symmetric, i.e.,

$$(\tilde{Q}_i^{(l+1)}, m_i^{(l)}) = \mathcal{H}_{\text{rfp}}^{(l)}(L_g[P_i^{(l)}], L_g[\tilde{Q}_i^{(l)}], E_i). \quad (29)$$

This is because

$$\begin{aligned}
L_g [m_{ij}^{(l)}] &= \phi_e^{(l)} \left(L_g [\tilde{Q}_i^{(l)}], L_g [\tilde{Q}_j^{(l)}], \|L_g [P_i^{(l)}] \right. \\
&\quad \left. - L_g [P_j^{(l)}]\|^2 \right) \\
&= \phi_e^{(l)} \left(\tilde{Q}_i^{(l)}, \tilde{Q}_j^{(l)}, \|L_g [P_i^{(l)} - P_j^{(l)}]\|^2 \right) \\
&= \phi_e^{(l)} \left(\tilde{Q}_i^{(l)}, \tilde{Q}_j^{(l)}, \|P_i^{(l)} - P_j^{(l)}\|^2 \right) = m_{ij}^{(l)}.
\end{aligned}$$

The derivation holds because the relative squared distance between two absolute spatial vectors does not change with L_g . Also, we have

$$\begin{aligned}
L_g [\tilde{Q}_i^{(l+1)}] &= \phi_q^{(l)} \left(L_g [\tilde{Q}_i^{(l)}], \sum_{j \in \mathcal{N}(i)} L_g [m_{ij}^{(l)}] \right) \\
&= \phi_q^{(l)} \left(\tilde{Q}_i^{(l)}, \sum_{j \in \mathcal{N}(i)} m_{ij}^{(l)} \right) = \tilde{Q}_i^{(l+1)}.
\end{aligned}$$

Therefore, we have (29) satisfied and $\mathcal{H}_{\text{rfp}}^{(l)}$ is invariantly symmetric. Now, we move forward to show that the SFP sub-module $\mathcal{H}_{\text{sfp}}^{(l)}$ is equivariantly symmetric, i.e.,

$$L_g [P_i^{(l+1)}] = \mathcal{H}_{\text{sfp}}^{(l)} \left(L_g [P_i^{(l)}], L_g [\tilde{Q}_i^{(l)}], E_i, L_g [m_{ij}^{(l)}] \right). \quad (30)$$

We apply the transformation L_g to the $\mathcal{H}_{\text{sfp}}^{(l)}$ in (22):

$$\begin{aligned}
&\phi_p^{(l)} \left(L_g [\tilde{Q}_i^{(l)}] \right) \cdot L_g [P_i^{(l)}] \\
&+ C \sum_{j \in \mathcal{N}(i)} \phi_m^{(l)} \left(L_g [m_{ij}^{(l)}] \right) \cdot \left(L_g [P_i^{(l)}] - L_g [P_j^{(l)}] \right) \\
&= \phi_p^{(l)} \left(\tilde{Q}_i^{(l)} \right) \cdot L_g [P_i^{(l)}] + C \sum_{j \in \mathcal{N}(i)} \phi_m^{(l)} \left(m_{ij}^{(l)} \right) \\
&\quad \cdot L_g [P_i^{(l)} - P_j^{(l)}] \\
&= L_g \left[\phi_p^{(l)} \left(\tilde{Q}_i^{(l)} \right) \cdot P_i^{(l)} + C \sum_{j \in \mathcal{N}(i)} \phi_m^{(l)} \left(m_{ij}^{(l)} \right) \right. \\
&\quad \left. \cdot (P_i^{(l)} - P_j^{(l)}) \right] \\
&= L_g [P_i^{(l+1)}].
\end{aligned}$$

Therefore, we have (30) satisfied. Combining both conditions (29) and (30), the equivariant symmetry in $P_i^{(l)}$ and the invariant symmetry in $Q_i^{(l)}$ are inherited through SIM layers in SiGNN, and thus, (28) is satisfied. As the sufficient condition (28) is met, (27) holds and the equivariant symmetry of policy $\pi_{\text{SiGNN},i}$ has been proved. $\square$

The second claim regarding the SiGNN-based value network states as:

Proposition 2: The SiGNN-based value network $V = \psi_{\text{SiGNN}}(o, J; \psi)$ is invariantly symmetric under the rotation transformation L_g , which satisfies:

$$V = \psi_{\text{SiGNN}}(L_g[o], L_g[J]; \psi). \quad (31)$$

Proof: The all-one matrix J is a constant matrix which is invariant to L_g . Also, Q is invariant. Thus,

$$V = \psi_{\text{SiGNN}}(L_g[o], L_g[J]; \psi) = \psi_{\text{SiGNN}}(L_g[P], Q, J; \psi).$$

Inside the value network ψ_{SiGNN} , the SiGNN releases $L_g[P_i^{(L)}]$ via (24) and feeds it to (25) to calculate the hidden embedding U_i using $L_g[\mathbf{o}_i^{\text{vel}(L)}]$ in $L_g[P_i^{(L)}]$. The hidden embedding U_i is invariant to L_g because:

$$L_g[U_i] = \|L_g[\mathbf{o}_i^{\text{vel}(L)}]\|^2 = \|\mathbf{o}_i^{\text{vel}(L)}\|^2 = U_i.$$

Note that the squared norm of a vector is equivalent to the relative squared distance between itself and the origin vector, which does not change with L_g . As a result, $L_g[U] = U$. Then,

$$\begin{aligned}
\psi_{\text{SiGNN}}(L_g[P], Q, J; \psi) &= \text{FC}(\tanh(L_g[U], Q); \psi_2) \\
&= \text{FC}(\tanh(U, Q); \psi_2) = V.
\end{aligned}$$

The proposition 2 has been proved. $\square$

VI. EXPERIMENTS AND EVALUATION

This section presents the experiments conducted to evaluate the performance of the proposed SiGNN-based MARL method. Experimental results in comparison with baselines are discussed in detail. Supplementary is available at <https://rongyeshi.github.io/SiGNNSupplementary.pdf>.

A. Experimental Settings

We implement a simulation environment using Python for the UAV swarm communication coverage task on a workstation equipped with 1 NVIDIA 4090 GPU, 1 Intel Core i9-12900KF @3.50 GHz CPU and Ubuntu 18.04 operating system. The control models are implemented and trained using PyTorch 1.3.0. We set the continuous target area as a 2D plane with $L_{\text{length}} = L_{\text{length}} = 100$ units. In each episode, 30 PoIs are sampled from a mixture of three Gaussian distributions, whose means are uniformly generated from the target area. We employ the parameter-sharing technique to train a swarm with $\{5, 10, 15, 20\}$ UAVs in the training phase to examine the scalability. We consider both global and partial observation scenarios in the experiments, where the partial observation involves $R_{\text{obs}} \in \{30, 50, 70\}$. The coverage radius is $R_{\text{cov}} = 8$. We set $e_0, \beta_{\text{speed}}, \beta_{\text{acc}} = \{1.1, 0.1, 0.05\}$.

By default, the proposed symmetry-informed MARL integrates the SiGNN to the MAPPO framework. The implementation details are provided in Table II. Specifically, we train the models one million steps (i.e., 50,000 episodes). The CAE scores and expected accumulated rewards (joint returns) are calculated every 100 episodes, with 5 test rollouts. We also record their coverage, energy and overlap breakdowns to facilitate a comprehensive evaluation of the proposed method's performance and comparison with baseline methods.

TABLE II
COEFFICIENTS AND HYPERPARAMETERS IN EXPERIMENTS

Coefficients and Hyperparameters	value
episode length T	200
$\lambda^{\text{cov}}, \lambda_1^{\text{eg}}, \lambda_2^{\text{eg}}, \lambda^{\text{ol}}$	$\{15, 0.05, 0.1, 10\}$
total training steps	10,000,000
MLP hidden size	[128,128]
activation function	ReLU
discount factor γ	0.99
critic loss	huber loss
huber delta	10
optimizer	Adam
optimizer epsilon	1e-5
network initialization	orthogonal
number of SIM layers L	2
gradient clip parameter	0.2
actor/critic learning rate	0.0002
buffer length	2000

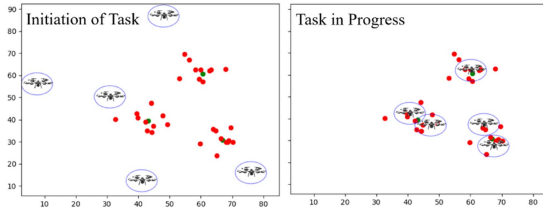


Fig. 4. Visualization of the UAV positions and dynamics in the simulation.

Fig. 4 illustrates UAV positioning and dynamics in the 2D simulation for the communication coverage task with 5 UAVs. The left part shows the initial UAV positions, while the right part displays the UAVs moving to cover PoIs under a control policy. The UAV swarm aims to achieve a high CAE score in serving the PoIs.

B. Learning-Based and Non-Learning-Based Baselines

We benchmark our SiGNN-based model against a range of baseline methods with classic and state-of-the-art network architectures, including:

- 1) *GraphSAGE*: GraphSAGE [60] is a straightforward yet robust graph neural network, extensively applied in tasks like social network analysis and traffic management using graph neural networks. It generates node embeddings by iteratively aggregating and transforming feature information from fixed-size local neighborhoods using trainable neural networks. Because the interactions between UAVs can be modeled as a graph, GraphSAGE is well-suited to swarm control tasks.
- 2) *MLP*: Multi-Layer Perceptrons (MLPs) are widely used in various standard algorithms within the MARL framework. The comparison with MLPs is intended to emphasize our benefits over traditional baseline algorithms.
- 3) *RNN*: Recurrent Neural Networks (RNNs) possess a structure that models temporal information and are widely employed in MARL to tackle problems of partial observability. We included RNNs in the comparisons to highlight the effectiveness of our method in scenarios with partial observations.

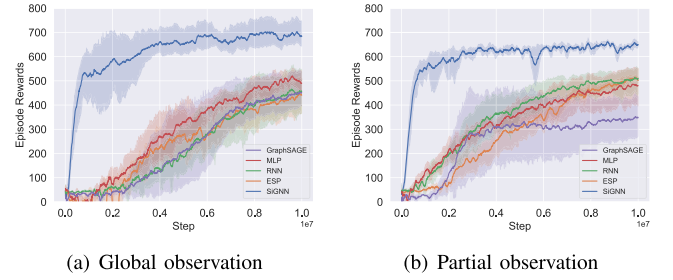


Fig. 5. Curves of expected episode rewards for global observation scenario and partial observation scenario.

- 4) *ESP*: A recent innovation, Exploitation of Symmetry Prior (ESP) method incorporates symmetry into the MARL training with soft regularization through data augmentation and specialized design of loss functions [48]. Our comparison with ESP aims to demonstrate the superior performance of our method.

Additionally, we also make comparison against three non-learning baselines:

- 1) *Random*: The random policy controls each UAV to select an action randomly at each time step t .
- 2) *MB-Greedy*: The Model-Based Greedy (MB-Greedy) method attempts to identify the action that maximizes the reward r_t^t within all possible actions in the simulated environment. Because the simulator is required to calculate an action, it is a model-based method. Unlike the typical approach, we discretize the continuous action space of $[-v_{\max}, v_{\max}]$ into ten values for sampling.
- 3) *MB-GA*: Genetic Algorithm (GA) is employed to approximate the optimal joint action a^t at each timestep t that maximizes the total return $\sum_t (\gamma)^t r(s^t, a^t)$. It iterates selection, crossover, and mutation on a population of candidate actions. This strategy also relies on an environment model and is model-based. Similar to MB-Greedy, we discretize the continuous action space.

C. Convergence and Performance on Global Observation

The proposed symmetry-informed MARL integrates SiGNN as the core component. As a learning-based method, it is important to evaluate the convergence efficiency in the training phase. The expected episode reward is the direct metric to evaluate the training stability and convergence. We first discuss the global observation case, i.e., $R_{\text{obs}} = 100$. The curves of expected episode rewards are showed in Fig. 5(a).

From Fig. 5(a), it is observed that SiGNN exhibits a faster, superior and more stable convergence than other learning-based algorithms. Specifically, SiGNN achieves high rewards above 650 and stabilizes there after 0.4 million training steps. Other baselines need about 0.9 million steps to stabilize at around the rewards 400~500.

Regardless of the superior performance of SiGNN in episode rewards, another concern arises from the fact that the reward function is adjusted from the metrics to decompose the individual influences of each UAV to enlarge the subtle difference

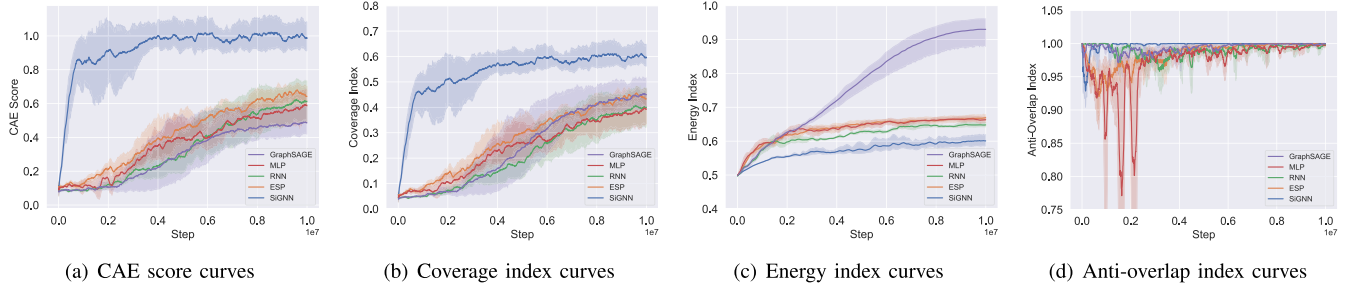


Fig. 6. The curves of CAE scores and corresponding indexes for scenario with global observation.

among sub-optimal policies during training. Therefore, a high reward does not guarantee the performance with respect to CAE (coverage, anti-overlap, energy). Consequently, it is necessary to examine the performance in terms of CAE scores and corresponding component indexes.

The experimental results are presented in Fig. 6. The CAE scores in Fig. 6(a) show that the SiGNN-based method significantly outperforms the baselines. Although there are slight changes in the final CAE scores' ranking, the overall curve trends are similar to the episode reward performance in Fig. 5(a) and the SiGNN-based method consistently and significantly outperforms the baselines. These findings confirm that the designed reward function positively correlates with the CAE score, and the distinction becomes more clear when using the reward metric (at a scale of 10^2). Among baselines, the ESP method leverages the symmetry prior in a soft constrained manner, achieving a better CAE score curve.

Fig. 6(b), (c), and (d) display the coverage, energy, and anti-overlap indexes, respectively, revealing more refined details in training. Specifically, regarding the coverage index, all methods incrementally learn to effectively serve the PoIs. The energy index of the curves consistently increases because of the energy consumption in an episode, and a flattened energy curve suggests that the UAVs maintain their positions without abrupt movements. The anti-overlap index curves exhibit an interesting pattern: At the beginning, the curves start from one as all UAVs are randomly distributed and no overlapping occurs. When the training takes effects, some methods tend to gather with each other and the anti-overlap index is negatively effected. Finally, as the UAVs are penalized by the overlapping, they learn to avoid redundant serving, and the index converges back to value one.

A closer look into the curves reveals that the GraphSAGE method performs well in coverage index and anti-overlap index due to its ability to effectively capture the spatial relationships among UAVs using graph-processing techniques. However, it fails to learn to hover while serving PoIs, leading to a deterioration in the energy index. The ESP, RNN, and MLP need to learn to recognize the environmental symmetry and then to avoid overlapping, which results in initial declining trends in the anti-overlap index during the early stages of training. In contrast, SiGNN quickly learns to provide communication coverage with greater energy efficiency. Although some initial overlap occurs during the initial stages of training, SiGNN adapts

to avoid redundant coverage faster than baselines. This might be attributed to the fact that SiGNN does not need to learn symmetry, allowing itself to concentrate more on enhancing energy performance during training.

D. Convergence and Performance on Partial Observation

Next, we turn to the more practical scenarios where each UAV can access to partial observation. We exemplify the discussion using $R_{\text{obs}} = 50$. Scenarios with other radii exhibit similar patterns, which are shown in supplementary Section I.

As shown in Fig. 5(b), under the partial observation, the GraphSAGE method performs the worst (i.e., the lowest reward and largest variance) among the baselines, probably because it adheres to local information processing and fails to learn cooperative behaviors. This disadvantage might be mitigated by using multi-hop techniques. The RNN method benefits more from the setting, because it is able to process sequential and temporal observation effectively, integrating temporal context and past events in memory to learn the symmetry property to coordinate the UAV swarm. The SiGNN-based method still maintains the highest training efficiency, demonstrating its ability to overcome the disadvantage of local observation. Compared with the global scenario in Fig. 5(a), the partial observation setting presents challenges to cooperation, resulting in relatively spiky learning curves and generally lower rewards.

Similar trends are also displayed in CAE scores and corresponding indexes in Fig. 7, where by limiting the observable information, the overall CAE scores reduce, and the index curves are generally noisy. The GraphSAGE's performance deteriorates more sharply compared to other baseline models. The proposed SiGNN-based method continues to excel in terms of CAE score and the three indexes.

In both global and partial observation settings shown in Fig. 5(a) and (b), by integrating the rotation symmetry into the value/policy networks' structure, the SiGNN-based method presents significantly improved efficiency compared to those without using symmetry-preserving networks (e.g., MLP, RNN). And this echoes the fact that, thanks to the symmetry-preserving networks, there is no need for additional trial-and-error to learn the symmetry property within the symmetric Dec-POMDP, resulting in notably higher training efficiency.

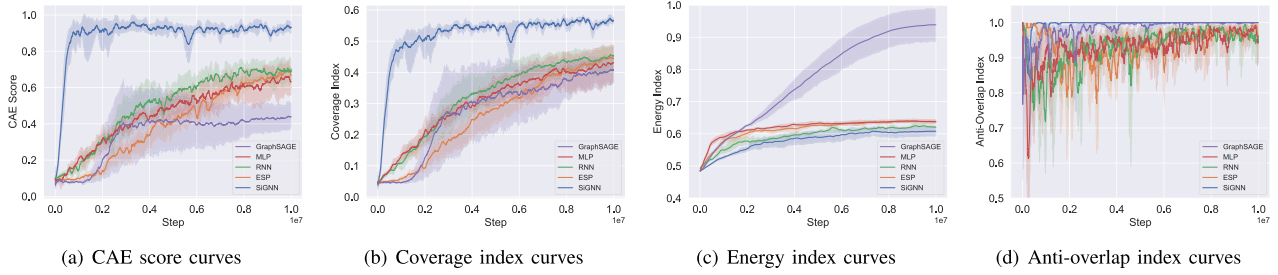


Fig. 7. The curves of CAE scores and corresponding indexes for scenario with partial observation.

TABLE III
THE METRIC PERFORMANCE OF NON-LEARNING AND LEARNING-BASED METHODS

Policy model	CAE Score	Coverage Index	Energy Index	Anti-Overlap Index
Random	0.1967 ± 0.1853	0.1560 ± 0.1433	0.7152 ± 0.1248	0.8511 ± 0.1133
MB-Greedy	0.3426 ± 0.2238	0.1836 ± 0.0548	0.4956 ± 0.0873	0.9247 ± 0.0344
MB-GA	0.5475 ± 0.1673	0.3329 ± 0.0732	0.5784 ± 0.0624	0.9513 ± 0.0233
GraphSAGE	0.4379 ± 0.0893	0.4065 ± 0.0537	0.9387 ± 0.0645	0.9985 ± 0.0015
MLP	0.6313 ± 0.0936	0.4304 ± 0.0752	0.6382 ± 0.0084	0.9420 ± 0.0673
RNN	0.7017 ± 0.0810	0.4529 ± 0.0528	0.6208 ± 0.0047	0.9635 ± 0.0528
ESP	0.6663 ± 0.0471	0.4439 ± 0.0142	0.6373 ± 0.0053	0.9572 ± 0.0691
SiGNN	0.9313 ± 0.0376	0.5655 ± 0.0109	0.6078 ± 0.0181	1.0000 ± 0.0000

The best model is indicated by bold. More results are available in supplementary Sec II.

E. Learning-Based Versus Non-Learning-Based Approaches

We compare the learning-based and non-learning-based approaches under the partial observation setting. The learning-based approaches are trained through 1 million steps. Each method are measured using 10 test episodes and we use Table III to display the CAE scores and the three indexes.

The Random method selects actions arbitrarily, resulting in highly unstable performance characterized by large variance and the poorest metric scores. The MB-Greedy exploits the simulator to determine the one-step action with maximal immediate reward. However, it fails to fully capture the cooperative relationships among UAVs which involves planning in multiple steps, the final performance of MB-greedy is limited. The MB-GA method demonstrates clear advantages over the other two traditional methods. It leverages genetic evolution heuristics to find near-optimal joint actions, performing well with CAE scores around 0.5. The two model-based methods give the best energy performance, probably because they tend to stay still after reaching one of the the PoIs. However, it struggles to achieve global optimality and the computation load is high, preventing it from real-time application.

The learning-based methods generally perform better than the non-learning-based baselines (generally with CAE scores > 0.5). They can conduct trial-and-errors to model the environment and learn to cooperate among UAVs, achieving more comprehensive and efficient strategy. Remarkably, the SiGNN-based method consistently presents higher and more stable performance in the CAE score and most indexes, underscoring the crucial role that the symmetry-informed network structure plays in learning optimal policies. Its energy performance is also the best among learning-based methods.

To highlight the advantages of the SiGNN-based method, we employ a histogram (see Fig. 8) to illustrate the percentage improvements in CAE scores with respect to the MB-Greedy

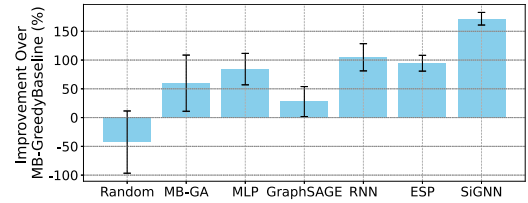


Fig. 8. Percentage Improvements over MB-Greedy Baseline.

baseline. In this representation, blue rectangles indicate the average improvements and black lines depict the variances. It is observable that, except for the Random algorithm, all tested algorithms exhibit improvements in performance compared to the MB-Greedy algorithm. Particularly notable is the SiGNN-based method, which not only achieves the highest improvement but also demonstrates consistent superior performance, highlighting the robustness of the algorithm.

F. Sensitivity to Different Hyperparameters

We study the sensitivity of the SiGNN-based method to different hyperparameters, including the MLP hidden size, learning rate, and the number of SiGNN layers L .

For the MLP hidden size, shown in Fig. 9(a), reducing the size to 1/4 of the default configuration (i.e., [32, 32]) results in a slower learning curve, implying that an overly simplified model may experience reduced performance. Increasing the size to 4 times (i.e., [512, 512]) does not bring significant performance improvement compared to the default size.

As shown in Fig. 9(b), the learning rate lr plays an important role in achieving high training efficiency and overall performance. The curve with an overly small learning rate ($lr = 0.00002$) grows steadily but stabilizes at a lower reward level, whereas the curve with an overly large learning rate ($lr = 0.002$)

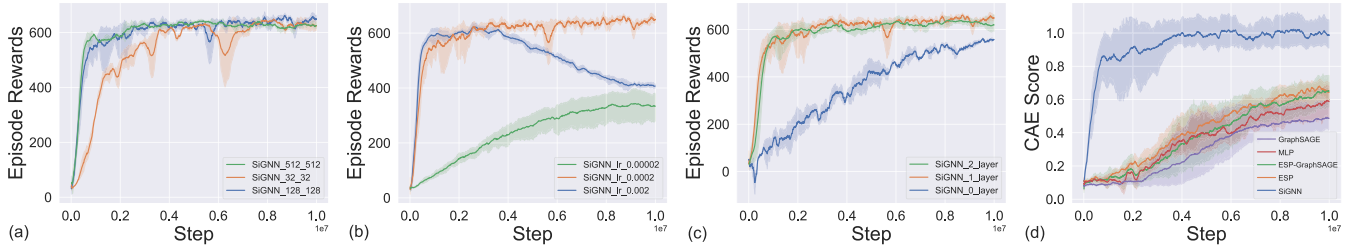


Fig. 9. Left: Performance under different hyperparameters, including (a) MLP hidden size, (b) learning rate, (c) SiGNN layer number. Right: (d) Comparison between rotational data augmentation-based methods and SiGNN-based method.

exhibits faster initial progress but fails to achieve high reward levels, indicating a deteriorated convergence.

Finally, as shown in Fig. 9(c), a significant improvement in training efficiency and performance is achieved by introducing Symmetry-Informed Module with $L = 1$, compared to the $L = 0$ configuration (correspond to the standard MAPPO). Further increasing L to 2 does not yield significant performance improvements; however, a smoother learning curve is observed, suggesting that additional SiGNN layers may enhance the symmetry property and the training stability.

G. Rotational Data Augmentation Versus SiGNN

Rotational symmetry is considered a basis of data augmentation in reinforcement learning, which can expand the original small amount of data. In Fig. 9(d), we compare the performance of data augmentation-based methods with the SiGNN-based method. Specifically, the ESP method proposed in [48] integrates rotational symmetry-based data augmentation into a backbone model (i.e., MLP) to enhance performance. To provide a more comprehensive analysis, we also apply the data augmentation technique to the GraphSAGE method, forming the ESP-GraphSAGE baseline for comparison.

Experimental results indicate that both the ESP and ESP-GraphSAGE methods consistently outperform their respective backbone models, MLP and GraphSAGE. This improvement is attributed to the data augmentation technique, which provides additional trial-and-error samples to learn the symmetry property in the UAV swarm environment, thereby enhancing traditional schemes. In contrast, the SiGNN method significantly outperforms the data augmentation-based approaches. This superiority stems from the intrinsic embedding of symmetry within the network structure, eliminating the need for additional trial-and-error processes to learn the symmetry property, and ultimately resulting in superior training efficiency.

H. Robustness Against Communication Drop

This subsection examines the robustness of SiGNN in scenarios where communication connections and information sharing among UAVs may fail with different probabilities.

In real world, besides partial observability, we should realize that various factors can disrupt communication among UAVs and it is more practical to assume that information sharing could be interrupted with a dropping probability p . In this experiment, we

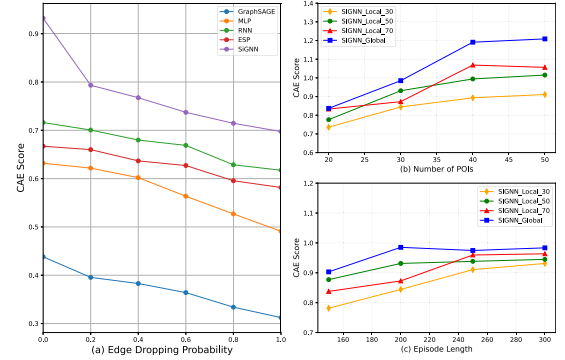


Fig. 10. CAE scores under different dropping rates, PoI numbers, and episode lengths.

set $p = \{0, 0.2, 0.4, \dots, 1.0\}$ and test the CAE performance of SiGNN and baselines.

The experimental results are showed in Fig. 10(a), from which we observe a consistent decline in CAE with increasing drop rates, indicating that reduced connectivity and information sharing adversely affect inter-UAV cooperation. An interesting phenomenon is that all the learning-based methods are still able to effectively perform coverage tasks even under $p = 1$ (i.e., no information sharing among UAVs), reaching a proper CAE above 0.5 (except for GraphSAGE).

Although the performance of the SiGNN-based method significantly deteriorates as p increases to 0.2, it stabilizes thereafter and consistently outperforms all other learning-based methods, demonstrating the robustness of the proposed SiGNN. Our insight into this phenomenon is that the communication between UAVs is inherently symmetric. Consequently, the information missing from some adjacent UAVs can be compensated and inferred from other connectable UAVs.

I. Performance Under Different PoIs and Episode Lengths

For further evaluation, the proposed SiGNN-based method is tested under different numbers of PoIs and episode lengths. With 5 UAVs, the number of PoIs is increased from 20 to 50. As shown in Fig. 10(b), the CAE scores generally show an upward trend as the number of PoIs increases. This is because a higher density of PoIs makes it more convenient for UAVs to locate and serve users. However, the CAE scores begin to flatten between 40 and

TABLE IV
SCALABILITY PERFORMANCE UNDER PARTIAL OBSERVATION

Number	Steps	SiGNN	MLP	RNN	ESP	GraphSAGE
5	300k	619.76 ± 26.43	286.22 ± 32.60	331.68 ± 44.21	198.59 ± 29.30	288.24 ± 38.50
	1000k	649.61 ± 26.11	489.38 ± 52.24	517.68 ± 29.90	501.34 ± 34.14	364.69 ± 54.24
10	300k	378.06 ± 23.45	39.84 ± 38.26	53.41 ± 38.89	-6.62 ± 51.92	-6.57 ± 54.14
	1000k	462.59 ± 33.42	75.71 ± 55.96	90.77 ± 49.17	83.47 ± 48.53	-235.17 ± 78.42
15	300k	235.31 ± 31.50	-202.66 ± 41.68	-214.55 ± 44.50	-104.39 ± 67.43	-144.95 ± 87.31
	1000k	226.83 ± 41.22	-58.44 ± 60.13	43.01 ± 69.55	55.68 ± 41.77	-526.73 ± 93.47
20	300k	50.68 ± 37.32	-322.46 ± 84.61	-336.59 ± 52.36	-426.25 ± 76.15	-558.18 ± 95.12
	1000k	122.14 ± 57.21	-378.82 ± 76.32	-78.14 ± 82.23	-118.54 ± 45.50	-943.23 ± 154.33

50 PoIs, indicating that the UAV swarm's coverage capacity may reach its limit due to the finite number of UAVs available.

A similar phenomenon can be observed as the episode length increases from 150 to 300, shown in Fig. 10(c). With short episode lengths (e.g., $T = 150$), the UAV swarm is still in the process of locating the PoIs, preventing efficient coverage. As the episode length increases, the policy has more time to coordinate the UAVs and maximize coverage. Eventually, the CAE scores plateau, as further improvements cannot be achieved simply by extending the episode time.

J. Scalability

This experiment was conducted to verify the scalability of various methods. The scalability refers to the ability of an algorithm to maintain high performance when the number of UAVs is increased.

In the simulation environment, we keep 30 PoIs constant and gradually increase the number of UAVs from 5 to 20. The performance of each method was evaluated in two phases of training: early stage at 300 k and later stage at 1000 k. This setup allows a comprehensive understanding of how SiGNN and other baseline methods perform under different training stages and scale conditions. Here, to make the the performance difference more visible, we choose the expected episode reward as the metric for comparison. The results are presented in Table IV.

All the methods show a trend of decreasing performance as the number of UAVs increases. For example, the MLP method significantly drops from 286 to -378. Despite the increased complexity and instability of larger swarms, the SiGNN-based method generalizes well and consistently achieves significantly higher reward values across various numbers of UAVs. In the task with 20 UAVs, the performance of all baseline models dropped to negative rewards, while SiGNN is the only model that consistently maintains its performance above 100. These results demonstrate the scalability advantages of the proposed method. More results are available in supplementary Section III.

K. Computational Overhead on Edge Computing Devices

In communication coverage tasks, MARL-based methods are used to control UAVs in a distributed manner. These methods involve expressing the communication protocols among UAVs through neural networks, which may require more computational resources than traditional RL. Such a requirement for increased resources brings up concerns regarding computational efficiency.

TABLE V
INFERENCE TIME ON EDGE COMPUTING DEVICES

Policy	Mean (s)	STD (s)	FPS
SiGNN_global	0.0243	0.0002	41
SiGNN_local	0.0239	0.0012	42
MLP_global	0.0125	0.0138	79
MLP_local	0.0144	0.0152	69
RNN_global	0.0227	0.0113	37
RNN_local	0.0231	0.0098	43
ESP_global	0.0138	0.0018	72
ESP_local	0.0151	0.0023	66
GraphSAGE_global	0.0279	0.0213	36
GraphSAGE_local	0.0286	0.0185	35

To demonstrate the practicality of our method on mobile computing devices, we implemented all the learning-based UAV swarm control models on the Jetson Nano, a device frequently utilized for controlling autonomous UAVs [61]. We evaluate their real-time performance on this platform to evaluate their real-world applicability. Each model executed 1000 steps to compute the average of inference times. The experimental results are displayed in Table V, including the inference times and frames per second (FPS) of different policy models on the Jetson Nano board. These experimental results reflect the real-time performance and computational complexity among different methods.

It is observed that the inference times of SiGNN-based series are very close across different observation distances, averaging between 0.0239 s and 0.0243 s, with a stable FPS of 41~42. The changes in local observation distance have little impact on inference time, indicating that observation distance is not the main bottleneck in the inference process. The SiGNN-based methods maintains high real-time performance, making it suitable for real-time UAV controlling tasks.

VII. CONCLUSION AND FUTURE WORK

This paper presents symmetry-informed MARL, leveraging a novel symmetry-informed graph neural network (SiGNN) as the policy and value networks. SiGNN embeds the inherent symmetry of multi-UAV systems into its structure, improving training efficiency for managing large swarms with continuous control. It includes an adjacency processing module to accommodate varying numbers of adjacent UAVs in partial observations. Theoretical analysis confirms that SiGNN preserves symmetry properties, ensuring the method's effectiveness. Simulation experiments for communication coverage show that SiGNN-based MARL surpasses advanced baselines in sample efficiency, scalability, and robustness.

In the future, we plan to extend the SiGNN-based framework to encompass a broader type of symmetry to handle more scenarios, such as swarms with heterogeneous UAVs and consider other types of multi-agent systems, such as combining UAVs with autonomous ground vehicles and robotic swarms for the communication coverage, disaster rescue and asteroid exploration tasks. Real-world testing and deployment of SiGNN-based method will be essential to validate its effectiveness outside of simulated environments and to ensure its practical utility in operational settings. Finally, how to properly share PoI information among UAVs to enhance performance while balancing computational and communication overhead is a promising topic for future research.

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